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use sys;
CREATE TABLE Employees (
EmpID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Dept VARCHAR(20) NOT NULL,
Role VARCHAR(20) NOT NULL,
Salary INT NOT NULL,
Gender CHAR(1) NOT NULL,
JoinDate DATE NOT NULL,
Address VARCHAR(100) NOT NULL,
City VARCHAR(50) NOT NULL,
State CHAR(2) NOT NULL,
Region VARCHAR(10) NOT NULL
);
INSERT INTO Employees (EmpID, Name, Dept, Role, Salary, Gender, JoinDate,
Address, City, State,
Region) VALUES
(1, 'Alice', 'HR', 'Manager', 55000, 'F', '2019-03-15', '101 Maple St', 'New
York', 'NY',
'East'),
(2, 'Bob', 'IT', 'Developer', 72000, 'M', '2020-07-22', '22 Oak Ave', 'Chicago',
'IL',
'Midwest'),
(3, 'Carol', 'Finance', 'Analyst', 68000, 'F', '2018-11-01', '303 Pine Rd', 'New
York', 'NY',
'East'),
(4, 'David', 'HR', 'Executive', 52000, 'M', '2021-01-10', '44 Birch Ln',
'Chicago', 'IL',
'Midwest'),
(5, 'Eve', 'IT', 'Senior Dev', 85000, 'F', '2017-06-30', '55 Cedar Blvd',
'Houston', 'TX',
'South'),
(6, 'Frank', 'Finance', 'Manager', 74000, 'M', '2019-09-14', '606 Elm St', 'New
York', 'NY',
'East'),
(7, 'Grace', 'Marketing', 'Executive', 60000, 'F', '2022-02-28', '77 Spruce Dr',
'Chicago', 'IL',
'Midwest'),
(8, 'Hank', 'IT', 'Lead', 90000, 'M', '2016-04-05', '88 Walnut Way', 'Houston',
'TX',
'South'),
(9, 'Ivy', 'HR', 'Executive', 58000, 'F', '2020-12-19', '99 Ash Ct', 'New York',
'NY',
'East'),
(10, 'Jack', 'Finance', 'Analyst', 65000, 'M', '2021-03-07', '10 Willow Pl',
'Chicago', 'IL',
'Midwest'),
(11, 'Karen', 'Marketing', 'Manager', 62000, 'F', '2018-08-23', '11 Poplar St',
'Houston', 'TX',
'South'),
(12, 'Leo', 'IT', 'Developer', 78000, 'M', '2019-11-11', '12 Sycamore Ave', 'New
York', 'NY',
'East'),
(13, 'Mia', 'HR', 'Executive', 53000, 'F', '2022-05-16', '13 Hickory Rd',
'Chicago', 'IL',
'Midwest'),
(14, 'Nate', 'Finance', 'Manager', 71000, 'M', '2017-12-02', '14 Chestnut Ln',
'Houston',
'TX', 'South'),
(15, 'Olivia', 'Marketing', 'Analyst', 66000, 'F', '2020-06-18', '15 Magnolia
Dr', 'New York',
'NY', 'East'),
(16, 'Paul', 'IT', 'Senior Dev', 82000, 'M', '2018-03-27', '16 Dogwood Blvd',
'Chicago', 'IL',

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'Midwest'),
(17, 'Quinn', 'HR', 'Executive', 57000, 'F', '2021-09-09', '17 Hawthorn St',
'Houston', 'TX',
'South'),
(18, 'Rita', 'Finance', 'Lead', 76000, 'F', '2019-01-25', '18 Redwood Ave', 'New
York', 'NY',
'East'),
(19, 'Sam', 'Marketing', 'Executive', 63000, 'M', '2022-07-14', '19 Cypress Ct',
'Chicago', 'IL',
'Midwest'),
(20, 'Tina', 'IT', 'Lead', 88000, 'F', '2016-10-31', '20 Juniper Way',
'Houston', 'TX',
'South'),
(21, 'Uma', 'HR', 'Manager', 60000, 'F', '2020-04-03', '21 Aspen Pl', 'Dallas',
'TX',
'South'),
(22, 'Victor', 'Finance', 'Analyst', 69000, 'M', '2018-07-19', '22 Cottonwood
St', 'Dallas', 'TX',
'South'),
(23, 'Wendy', 'Marketing', 'Manager', 64000, 'F', '2021-11-27', '23 Palmetto
Ave', 'Phoenix',
'AZ', 'West'),
(24, 'Xander', 'IT', 'Architect', 95000, 'M', '2015-02-14', '24 Mesquite Rd',
'Phoenix', 'AZ',
'West'),
(25, 'Yara', 'HR', 'Executive', 54000, 'F', '2022-09-08', '25 Saguaro Ln',
'Phoenix', 'AZ',
'West'),
(26, 'Zoe', 'Finance', 'Manager', 73000, 'F', '2019-05-21', '26 Palo Verde Dr',
'Dallas', 'TX',
'South'),
(27, 'Aaron', 'Marketing', 'Analyst', 67000, 'M', '2020-10-15', '27 Ocotillo
Blvd', 'Phoenix',
'AZ', 'West'),
(28, 'Bella', 'IT', 'Senior Dev', 80000, 'F', '2017-08-06', '28 Ironwood St',
'Chicago', 'IL',
'Midwest'),
(29, 'Carlos', 'HR', 'Executive', 56000, 'M', '2021-06-24', '29 Pecan Ave',
'Houston', 'TX',
'South'),
(30, 'Diana', 'Finance', 'Lead', 77000, 'F', '2018-02-09', '30 Peachtree Ct',
'New York', 'NY',
'East'),
(31, 'Ethan', 'IT', 'Developer', 74000, 'M', '2019-04-18', '31 Magnolia St',
'Dallas', 'TX',
'South'),
(32, 'Fiona', 'Marketing', 'Executive', 61000, 'F', '2022-01-05', '32 Bluebonnet
Ave', 'Dallas',
'TX', 'South'),
(33, 'George', 'Finance', 'Analyst', 66000, 'M', '2020-08-11', '33 Sunflower
Rd', 'Phoenix',
'AZ', 'West'),
(34, 'Helen', 'HR', 'Manager', 59000, 'F', '2017-03-22', '34 Cactus Ln',
'Phoenix', 'AZ',
'West'),
(35, 'Ivan', 'IT', 'Lead', 92000, 'M', '2015-09-30', '35 Desert Dr', 'Phoenix',
'AZ',
'West'),
(36, 'Julia', 'Marketing', 'Manager', 65000, 'F', '2019-12-03', '36 Mesa Blvd',
'Dallas', 'TX',
'South'),
(37, 'Kevin', 'Finance', 'Manager', 70000, 'M', '2021-07-20', '37 Prairie St',
'Chicago', 'IL',
'Midwest'),

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(38, 'Laura', 'HR', 'Executive', 55000, 'F', '2022-10-14', '38 Lakeview Ave',
'Chicago', 'IL',
'Midwest'),
(39, 'Mark', 'IT', 'Developer', 76000, 'M', '2018-05-07', '39 Riverview Rd',
'New York', 'NY',
'East'),
(40, 'Nina', 'Marketing', 'Analyst', 63000, 'F', '2020-03-25', '40 Bayview Ln',
'New York',
'NY', 'East'),
(41, 'Oscar', 'Finance', 'Analyst', 67000, 'M', '2019-08-19', '41 Hillside Dr',
'Dallas', 'TX',
'South'),
(42, 'Priya', 'HR', 'Senior Dev', 79000, 'F', '2016-11-28', '42 Creekside Blvd',
'Houston', 'TX',
'South'),
(43, 'Raj', 'IT', 'Manager', 85000, 'M', '2018-06-14', '43 Brookside St',
'Phoenix', 'AZ',
'West'),
(44, 'Sarah', 'Marketing', 'Lead', 71000, 'F', '2017-02-01', '44 Springdale
Ave', 'New York',
'NY', 'East'),
(45, 'Tom', 'Finance', 'Senior Dev', 81000, 'M', '2020-09-09', '45 Fairview Rd',
'Chicago', 'IL',
'Midwest'),
(46, 'Uma', 'IT', 'Architect', 97000, 'F', '2014-12-22', '46 Northgate Ln', 'New
York', 'NY',
'East'),
(47, 'Vera', 'Marketing', 'Manager', 68000, 'F', '2021-04-30', '47 Eastwood Dr',
'Dallas', 'TX',
'South'),
(48, 'Will', 'HR', 'Analyst', 51000, 'M', '2022-11-15', '48 Westfield Blvd',
'Phoenix', 'AZ',
'West'),
(49, 'Xena', 'Finance', 'Lead', 78000, 'F', '2019-10-06', '49 Southpark St',
'Chicago', 'IL',
'Midwest'),
(50, 'Yusuf', 'IT', 'Senior Dev', 87000, 'M', '2016-07-17', '50 Central Ave',
'Houston', 'TX',
'South');

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Select * from employees;

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-- q1

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select dept, count(EmpId) as 'TotalSalary' from employees group by dept;

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-- q2

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select dept, sum(salary) as 'TotalEmployees' from employees group by dept;

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-- q3

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select gender, round(avg(Salary),2) as 'AvgSalary' from Employees group by
gender;

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-- q4

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select dept, max(salary) as 'MaxSalary' from employees group by dept;

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-- q5

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select role, min(salary) as 'MinSalary' from Employees group by role;

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-- q6

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select city, count(EmpId) as 'EmpCount' from Employees group by city;

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-- q7

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select region, sum(Salary) as 'TotalSalary' from Employees group by region;

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-- q8

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select dept, round(avg(salary),2) as 'AvgSalary' from Employees group by dept
order by AvgSalary desc;

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-- q9

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select city, sum(Salary) as 'TotalSalary' from Employees group by city order by
TotalSalary asc;

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-- q10

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select role, count(Empid) as 'EmpCount' from Employees group by role order by

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EmpCount desc,role asc;
-- q11
select state, count(EmpId) as 'EmpCount' from Employees group by state order by
EmpCount desc,state desc;
-- q12
select region, sum(salary) as 'TotalSalary' from Employees group by region order
by TotalSalary desc;
-- q13
select dept, min(Salary) as 'MinSalary' from Employees group by dept order by
MinSalary desc;
-- q14
select dept, count(Empid) as 'EmpCount' from Employees group by dept having
EmpCount>10;
-- q15
select dept, round(avg(salary),2) as 'AvgSalary' from Employees group by dept
having AvgSalary>=65000;
-- q16
select city, sum(Salary) as 'TotalSalary' from Employees group by city having
TotalSalary>=600000;
-- q17
select role, count(EmpId) as 'EmpCount' from Employees group by role having
EmpCount>7;
-- q18
select region, round(avg(salary),2) as 'AvgSalary' from Employees group by
region having AvgSalary<70000;
-- q19
select dept, max(Salary) as 'MaxSalary' from Employees group by dept having
MaxSalary>=90000;
-- q20
select state, count(EmpId) as 'EmpCount' from Employees group by state having
EmpCount>12;
-- q21
select city, min(Salary) as 'MinSalary' from Employees group by city having
MinSalary>=60000;
-- q22
select dept, city, count(EmpId) as 'EmpCount' from Employees group by dept,
city;
-- q23
select dept, gender, round(avg(salary),2) as 'AvgSalary' from Employees group by
dept,gender;
-- q24
select region, role, count(EmpId) as 'EmpCount' from Employees group by region,
role;
-- q25
select state, dept, max(Salary) as 'MaxSalary' from Employees group by state,
dept;
-- q26
select city, gender, sum(salary) as 'MaxSalary' from Employees group by city,
gender;
-- q27
select dept, city, count(EmpId) as 'EmpCount' from Employees group by dept, city
having EmpCount>2;
-- q28
select region, gender, round(avg(Salary),2) as 'AvgSalary' from Employees group
by region, gender
order by region desc, AvgSalary desc;
-- q29
select state, role, count(EmpId) as 'EmpCount' from Employees group by state,
role having EmpCount>1;
-- q30
select year(joindate) as 'JoinYear', count(EmpId) as 'Joiners' from Employees
group by Joinyear;
-- q31
select year(joindate) as 'JoinYear', round(avg(Salary),2) as 'AvgSalary' from

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Employees group by JoinYear;
-- q32
select month(joindate) as 'JoinMonth', count(EmpId) as 'Joiners' from Employees
group by JoinMonth;
-- q33
select quarter(joindate) as 'JoinQtr', sum(Salary) as 'TotalSalary' from
Employees group by JoinQtr;
-- q34
select quarter(joindate) as 'Qtr', count(EmpId) as 'Joiners' from Employees
group by Qtr having qtr in (1,3);
-- q35
select year(Joindate) as 'JoinYear', dept, count(EmpId) as 'EmpCount' from
Employees
group by dept,JoinYear
order by JoinYear asc, dept;
-- q36
select year(joindate) as 'JoinYear', count(EmpId) as 'Joiners' from Employees
group by JoinYear having Joiners>6;
-- q37
select quarter(joindate) as 'Qtr', round(avg(Salary),2) as 'AvgSalary' from
Employees group by Qtr having AvgSalary>70000;
-- q38
select state, count(EmpId) as 'EmpCount', round(Avg(Salary),2) as 'AvgSalary'
from Employees group by state;
-- q39
select state, dept, count(EmpId) as 'EmpCount' from Employees group by
state,dept;
-- q40 check
select city, count(dept) as 'DeptCount' from Employees group by city having
DeptCount>2;
-- q41
select region, state, sum(Salary) as 'TotalSalary' from Employees group by
region, state;
-- q42
select region, count(EmpId) as 'EmpCount', round(avg(Salary),2) as 'AvgSalary'
from Employees group by region having EmpCount>10 and AvgSalary>67000;
-- q43
select city, count(distinct(role)) as 'UniqueRoles' from Employees group by
city;
-- q44
select dept, count(EmpId) as 'FemaleCount' from Employees where gender='F' group
by dept;
-- q45
select dept, round(avg(Salary),2) as 'AvgSalary' from Employees where
joindate>'2019-01-01' group by dept;
-- q46
select city, sum(Salary) as 'TotalSalary' from Employees where dept='IT' group
by city;
-- q47
select role, count(EmpId) as 'EmpCount' from Employees where salary>70000 group
by role;
-- q48
select dept, max(Salary) as 'MaxSalary' from Employees where Region in ('South',
'West') group by dept;
-- q49
select year(Joindate) as 'JoinYear', count(EmpId) as 'EmpCount' from Employees
where state='TX' group by JoinYear;
-- q50
select dept, count(EmpId) as 'ElegibleFemales' from Employees where salary>60000
and gender='F' group by dept having ElegibleFemales>3;

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